

CORE COMPETENCIES

I. Biomedical Equipment Technology

Quality assurance procedure for ultrasound
Problems with and troubleshooting ultrasound
Nuclear medicine/gamma camera
Computed tomography (CT)
Positron emission tomography
Magnetic resonance imaging (MRI)
Radiation therapy
Computer applications in imaging
Medical network standards, DICOM, HL7
Picture archiving and communication system (PACS)
Power distribution system
Environmental control unit
Oxygen storage and generation system

D. Test Equipment and Procedures

Electrical safety analyzer
Electrical safety testing
Electrosurgical unit analyzer
Electrosurgical unit inspection
Medical ultrasound analyzer
Physiologic monitor inspection
Non-invasive blood pressure (NIBP) monitor analyzer
NIBP inspection
Invasive blood pressure inspection
Cardiac output monitor inspection
Pulse oximeter testing
Electrocardiograph simulator/analyzer
Electrocardiogram (ECG) testing
Telemetry inspection
Fetal monitor inspection
Infusion pump analyzer
Infusion pump inspection
Infant incubator inspection
Infant radiant warmer inspection
Defibrillator analyzer
Defibrillator testing
External pacemaker inspection
Heart-lung machine maintenance
Ventilator inspection and maintenance
Anesthesia machine maintenance
Gas analyzer
Pressure meter

Flowmeter
X-ray unit testing
Centrifuge preventive maintenance
Microscope maintenance
Water bath inspection
Slide stainer inspection
Water bath operation and maintenance
Laboratory incubator operation and maintenance
Slide stainer operation and maintenance
Diluter operation and maintenance
Cell counter inspection
Colorimeter operation and maintenance
Spectrophotometer operation and maintenance

E. Troubleshooting and Repair

Hospital grade plug
Schematic, wiring diagram
Circuit extraction
Relay troubleshooting
Ventilator operation and troubleshooting
Compressor operation and repair
Alternating pressure pump operation and troubleshooting
Heating pad pump operation and troubleshooting
Oxygen concentrator operation and troubleshooting
Low-volume suction machine operation and troubleshooting
Electrocardiograph operation and troubleshooting
Defibrillator troubleshooting
Physiologic monitor troubleshooting
NIBP troubleshooting
Invasive blood pressure troubleshooting
Pulse oximeter operation and troubleshooting
Infusion pump operation and troubleshooting
Fetal monitor operation and troubleshooting
Electrosurgical unit operation and troubleshooting
Sterilizer operation and troubleshooting
Diagnostic ultrasound troubleshooting
Dental x-ray operation and troubleshooting
X-ray unit operation and troubleshooting
Blood gas analyzer operation and troubleshooting
Centrifuge operation and troubleshooting
Centrifuge repair